



PHASE 1 — FOUNDATION

Electronics Workshop

Learn to read a circuit — the meter, switches, and symbols.

Say Watt Robotics · ROV Team | 2026-2027 MATE ROV Season

CLASS 1.4 · ONE SESSION · MADISON, ALABAMA

THE MEETING

Three hours, learning to read a circuit.

0:00	Welcome & Safety Brief	15 min
0:15	The Multimeter	35 min
0:50	How Switches Work	20 min
1:10	Simple Circuits Lab	30 min
1:40	<i>Break</i>	10 min
1:50	Build & Measure a Circuit	40 min
2:30	The Electrical Symbols Sheet	20 min
2:50	Wrap-up & homework	10 min

SECTION 01

Welcome & Safety Brief

0:00 - 0:15 · 15 MIN

01

Every meeting we ever hold will begin in exactly this place.

Every meeting opens with a safety brief.

Why we do this

Real engineering and dive teams brief safety before every job — not because something went wrong, but so it never does. We adopt the same habit on day one. Sixty seconds, every meeting, no exceptions.

Each brief answers the same four questions, so the team always knows what to expect.

- 1 What are we doing today?
- 2 What could hurt someone?
- 3 How do we prevent it?
- 4 Who do we tell if something happens?

TODAY'S BRIEF · CLASS 1.4

THE WORK

Multimeters and simple circuits — measuring and reading, not yet wiring the robot.

THE HAZARDS

Low-voltage circuits and batteries; the soldering iron if we tin a lead.

PREVENTION

Set the meter before connecting; never short a battery's terminals; irons in stands if used.

IF SOMETHING HAPPENS

Stop, disconnect the battery, tell the advisor, and log it.

LOOKING AHEAD

Class 1.5 — the control box build begins (the long one).

SECTION 02

The Multimeter

0:15 - 0:50 · 35 MIN

02

Before we trust a wire, we learn to read it.

Your window into the circuit.

A multimeter measures what your eyes can't: voltage (the push), continuity (is the path complete?), and current (the flow). With it, a kid can prove a wire is good before trusting it on the robot.

It is also how you find a fault. A dead thruster is usually an open circuit somewhere — a broken joint or a loose screw. The meter walks the path and finds the gap.

Measure, don't guess. The meter turns 'I think it's wired right' into 'I know it is.'

OHM'S LAW

Voltage = Current × Resistance.

$$V = I \times R$$

The one equation behind every reading. From the reading: Underwater Robotics, Ch 8 §4.4.

Three readings. Know what each tells you.

VOLTAGE

The push

How hard the electricity is being pushed, across two points. Our system runs at 12 volts.

CONTINUITY

A closed loop

Is the path complete? A beep means the circuit is closed. No beep means an open — a break to find.

CURRENT

The flow

How much is flowing, in amps. Too much means a short or overload — the reason we fuse the circuit.

Voltage, continuity, current — the three questions the meter answers.



SECTION 03

Switches & Circuits

0:50 - 1:40 · 50 MIN

03

A switch, a battery, a light — the whole idea of a circuit in your hands.

BEFORE CLASS

Watch it at home. Build it in class.

MATE's instructors teach the electronics on video. Watch these before class so our meeting time goes to measuring, wiring, and building — not slides.

As you watch, notice: how to set the meter's dial, how a switch makes and breaks a circuit, and how a simple circuit carries current from battery to load and back.

Come ready to measure and build — we pick up the meter at the door.

PRE-WATCH HOMEWORK

Three short videos, then straight to the bench.

[MATE: Multimeter Basics](#)

Plus MATE: How Switches Work and Simple Circuits — all assigned on the team site.

Six symbols you'll see all season.

1 Wire

A CONDUCTOR

A straight line carries current. A dot where wires connect; a hop where they cross without touching.

2 Battery

THE SOURCE

Long line positive, short line negative. The push that drives the circuit.

3 Switch

THE GATE

Opens and closes the path. SPST, SPDT, DPDT — the kinds you'll wire in the control box.

4 Resistor

LIMITS FLOW

A zigzag. Slows the current so a component gets just what it needs.

5 LED

A LIGHT

A diode that lights up — and only works one way around. Polarity matters.

6 Fuse

THE SAFETY

Breaks the circuit if current gets too high. Sized to the full-load amps.

Build a circuit that works.

- 1 Read the schematic**

Start from the symbols: battery, switch, resistor, LED. Know the path before you wire it.
- 2 Build it on the board**

Wire the Simple Circuits Lab Kit to match the schematic. One group, one circuit.
- 3 Flip the switch**

Close the circuit and the LED lights. Nothing? You have an open — walk it with the meter.
- 4 Measure and prove it**

Read the voltage across the LED and the continuity of each leg. Confirm it matches the plan.

Two things to remember. Every kid builds and measures one working circuit before we touch the control box.

Keep the symbols sheet. Forever.

The Electrical Symbols handout uses the same standard your competition SID will be graded against. The sheet you learn on today is the sheet you draw from in April.

MATE's Electrical Symbols handout is free and small, but it is one of the highest-leverage pages of the season. Print it, three-hole punch it, and keep it in the notebook all year.

The handout

MATE's one-page key to every schematic symbol. Free to print.

The SID

Your System Integration Diagram — a required, scored competition document drawn in these exact symbols.

Keep it forever

The standard doesn't change. Today's reference becomes April's grading rubric.

B R E A K

Ten minutes.

We resume in ten — meters down, hands clear of the bench.

SECTION 04

Wrap-Up & Homework

2:50 - 3:00 · 10 MIN

Name what happens next — and stage the parts before the holiday break.

04

WRAP-UP & HOMEWORK

Before we meet again.

EVERY STUDENT

- 1 Practice reading a simple schematic — battery, switch, resistor, LED.
- 2 Keep your Electrical Symbols handout in your notebook — you'll use it all year.
- 3 Software Lead: make sure the repo is ready for wiring notes.

THE PARALLEL TRACK · LEADS & ADVISORS

The control box is a long build (MATE budgets six hours) — stage the kit's control-box parts and the soldering stations. Battery in hand.

NEXT LESSON · CLASS 1.5

Control Box Part I

We build the PufferFish practice board, then start the TriggerFish REV 4 control box — the topside brain that drives the thrusters.



READ THE CIRCUIT

Now they can read it.

*Every kid can measure a circuit, read the symbols, and build a working one. The language of electronics is theirs.
Next, the control box.*